



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

MIDTERM EXAMINATION – SPRING 2024

COURSE NAME:

Computer Architecture

COURSE NUMBER:

SOC2060

EXAMINATION DATE:

TIME:

EXAMINATION DURATION:

90 Minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

None

SPECIAL INSTRUCTIONS:

- Answers will only be evaluated if they are readable.
- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

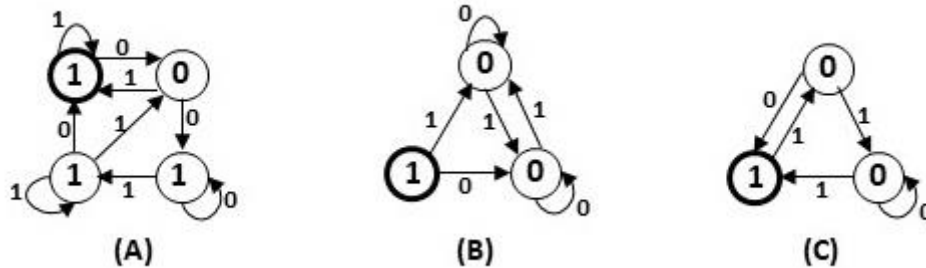
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1.[18 points] Finite State Machine

(a)[6 points] For each of the following FSMs, indicate if they are or are not well formed. Note that the state names have been omitted for clarity, you may assume the state names are unique.

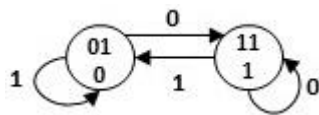


FSM A (circle one): **Not Well Formed**

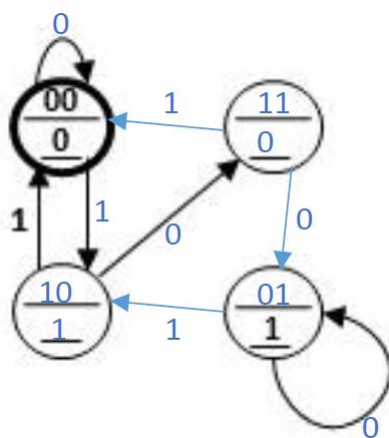
FSM B (circle one): **Well Formed**

FSM C (circle one): **Not Well Formed**

(b)[12 points] Given the partially completed truth table and FSM diagram below, complete all the missing entries in the truth table and the FSM diagram. In each state circle, the top entry is S_1S_0 and the bottom entry is the value of Out. Make sure that you have labeled all missing states, inputs and outputs, and that you have added and labeled any missing transitions in the FSM.



Note that in the above diagram, '01' and '11' in the circle are encoded states, and '0' and '1' in the circle are outputs. In state '01', if input is '0', next state is '11' and output is '0'. In state '01', if input is '1', next state is '01' and output is '0'.



S_1	S_0	In	S_1'	S_0'	Out
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	0	1	1
0	1	1	1	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1	0	0	0

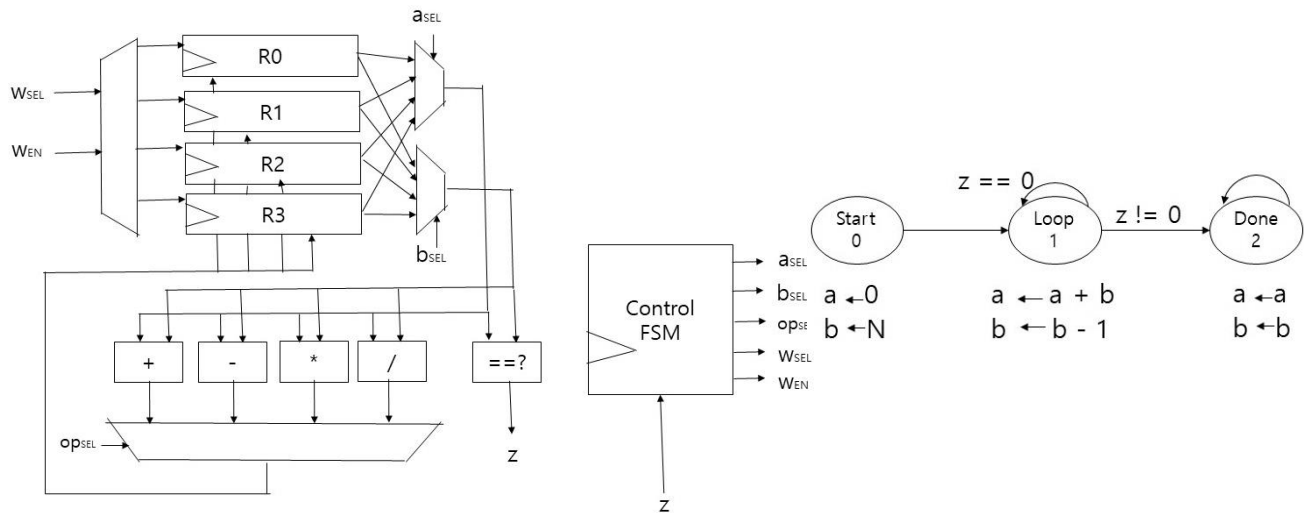
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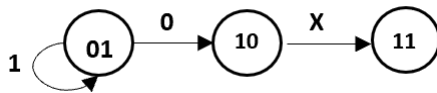
2.[20 points] Programmable Machine

(a)[20 points] Below is the picture of a programmable machine



Initially, R0 contains N, R1 contains k, R2 contains -1, R3 contains 1. In the final state, R3 has final result. Finish the FSM to calculate $N \cdot K$ (exponential). Start state is '00' and final state is '11'. k is a non-negative integer.

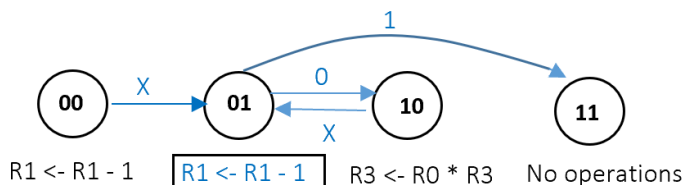
- Write the operation of state '01' inside the rectangle.
- Draw arrows, and write 'z' value (0 or 1) above each arrow. If 'z' value can be either 0 or 1, write 'X' above the arrow.



In the state '01' of the above example FSM, if $z == 0$, next state is '10', otherwise, next state is '01'.

In the state '10' of the above example FSM, next state is '11' regardless of the value of 'z'.

- Finish following tables



s	z	op _{SEL}	w _{SEL}	w _{EN}	a _{SEL}	b _{SEL}	s'
00	0	00	01	1	01	10	01
00	1	00	01	1	01	10	01

s	z	op _{SEL}	w _{SEL}	w _{EN}	a _{SEL}	b _{SEL}	s'
01	0	00	01	1	01	10	10
01	1	00	01	1	01	10	11

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(c)[3 points] Fill in the values of control signals in the STORE RESULT cycle of “LDR R3 R5 #9. In that cycle, the following operation is processed. $R3 \leftarrow MDR$

GateALU	GateMDR	LD.REG	LD.IR	LD.PC	LD.MAR	DR
0	1	1	0	0	0	011

4.[18 points] LC-3 Machine Language

LD (Load) 	ST (Store) 	LEA (Load Effective Address) 	BR (Conditional Branch)
LI (Load Indirect) 	STI (Store Indirect) 	ADD (Addition) 	AND (Bit-wise Logical AND)
LDR (Load Base + offset) 	STR (Store Base + offset) 	NOT (Load Base + offset) 	AND (Bit-wise Logical AND)

(a)[8 points] Finish LC-3 machine code to load R1 with integer 208 ($R1 \leftarrow 208$). Assume that your memory and registers are empty, and try to make it run as fast as possible. Write semantics of your instructions to the right of your instructions.

0	1	0	1	0	0	1	0	0	1	1	0	0	0	0	0	$R1 \leftarrow 0$
0	0	0	1	0	0	1	0	0	1	1	0	1	1	0	1	$R1 \leftarrow 13$
0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	1	$R1 \leftarrow R1 + R1; R1 \leftarrow 26$
0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	1	$R1 \leftarrow R1 + R1; R1 \leftarrow 52$
0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	1	$R1 \leftarrow R1 + R1; R1 \leftarrow 104$
0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	1	$R1 \leftarrow R1 + R1; R1 \leftarrow 208$

(b)[10 points] Count the number of ‘1’s in R1, and save the number in R0. If R1 has “01010101011111”, then R0 has 10. Write semantics of your instructions to the right of your instructions. Your instructions start at location x3000. Try to count the number of sign bits which are 1’s.

x3000	0	1	0	1	0	0	0	0	0	0	1	0	0	0	0	$R0 \leftarrow 0$
x3001	0	0	0	1	0	0	1	0	0	1	1	0	0	0	0	$R1 \leftarrow R1 + 0$
x3002	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	BR //skip next instruction
x3003	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	$R0 \leftarrow R0 + 1$
x3004	0	1	0	1	0	1	0	0	1	0	1	0	0	0	0	$R2 \leftarrow 0$
x3005	0	0	0	1	0	1	0	0	1	0	1	0	1	1	1	$R2 \leftarrow 15$ //set up counter
x3006	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	BR //branch out of loop
x3007	0	0	0	1	0	0	1	0	0	1	0	0	0	0	1	$R1 \leftarrow R1 + R1$
x3008	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	BR //skip next instruction
x3009	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	$R0 \leftarrow R0 + 1$
x300A	0	0	0	1	0	1	0	0	1	0	1	1	1	1	1	$R2 \leftarrow R2 - 1$
x300B	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	BR //branch to start of loop
x300C																

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5.[18 points] LC-3 Machine Language

The following is an LC-3 program which reads each array element from the file and compares each array element with the keyboard input from the user. The file contains ascii characters and ends with a special character of EOT (End of Text) which is x04. This program saves the total number of array elements which are the same as user's keyboard input into R2. The keyboard input is saved into R0 with ascii format by the operating system and R0 is written to the monitor by the operating system in ascii format. Memory[x3012] contains x3013, Memory[x3014] contains x04, and the user typed '0' with the keyboard.

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x3000	0	1	0	1	0	1	0	0	1	0	1	0	0	0	0	0
x3001	0	0	1	0	0	1	1	0	0	0	0	1	0	0	0	0
x3002	1	1	1	1	0	0	0	0	0	0	1	0	0	0	1	1
x3003	0	1	1	0	0	0	1	0	1	1	0	0	0	0	0	0
x3004	0	0	0	1	1	0	0	0	0	1	1	1	1	0	0	0
x3005	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0
x3006	1	0	0	1	0	0	1	0	0	1	1	1	1	1	1	1
x3007	0	0	0	1	0	0	1	0	0	1	1	0	0	0	0	1
x3008	0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	0
x3009	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	1
x300A	0	0	0	1	0	1	0	0	1	0	1	0	0	0	0	1
x300B	0	0	0	1	0	1	1	0	1	1	1	0	0	0	0	1
x300C	0	1	1	0	0	0	1	0	1	1	0	0	0	0	0	0
x300D	0	0	0	0	1	1	1	1	1	1	1	0	1	1	0	0
x300E	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0
x300F	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0
x3010	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	1
x3011	1	1	1	1	0	0	0	0	0	0	1	0	0	1	0	1
x3012	Starting address of file															
x3013	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0

R0 is written with the keyboard input

Display R0 into the monitor
Halt**(a)[3 points]** The PC has just become x3002 at the Fetch phase. What is the value of R2?

R2 = 0

(b)[3 points] The PC has just become x3012 at the Fetch phase. What is the value of R2?

R2 = 1

(c)[3 points] The PC has just become x3004 at the Fetch phase. What is the value of R0?

R0 = x0030

(d)[3 points] The PC has just become x3003 at the Fetch phase. What is the value of R3?

R3 = x3013

(e)[3 points] The PC has just become x3010 at the Fetch phase. What is the value of R3?

R0 = x0030

(f)[3 points] The PC has just become x3011 at the Fetch phase. What is the value of R0?

R0 = x0031

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6.[8 points] Karnough Map

(a)[8 points] Given a function F defined by the truth table to the right, complete the Karnough Map and find the lenient sum of products expression for F.

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	0	1
11	1	1	0	0
10	1	1	1	1

Minimum SOP expression for F:

$$\text{Inv}(D) + \text{Inv}(A) + \text{Inv}(B)\text{Inv}(C)$$

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

6.[9 points] ISA and ISA Trade-offs

(a)[9 points] Circle Uarch or ISA corresponding to whether each of the following characteristics is better classified as “microarchitecture” or “ISA”:

The speed of ALU's ADD operation: **Uarch**CPU endianness: **ISA**The number of pipeline stages in the CPU: **Uarch**The memory addressing modes available for arithmetic operations: **ISA**The program counter width: **ISA**Cache replacement policies: **Uarch**Cache associativity: **Uarch**Opcode of ADD instruction: **ISA**Condition codes: **ISA**